

RUPAK RAUT

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EDUCATION

The University of Southern Mississippi

Graduated: May 2026

Major: Computer Science & Mathematics

GPA: 4.0

Honors: Top 10% YC Applicant W24, Techstars Interview Finalist

Coursework: Data Structures & Algorithms, Databases, Operating Systems, Linear Algebra, Probability & Statistics

WORK EXPERIENCE

Software Engineer

Nov 2025 - Present

Retina Specialists of Mississippi

- Built and rolled out a HIPAA-compliant patient-intake and clinic-management platform across 7 practice locations, working on-site with clinical staff to scope requirements and drive each go-live.
- Integrated the Athena and Nextech EHRs over OAuth 2.0 to sync patient and appointment data, and built the patient-intake flow with QR-code sessions, dynamic forms, e-signatures, and client-side PDF generation.
- Engineered a type-safe PostgreSQL (Supabase) data layer using Drizzle ORM and Zod, enforcing 100+ row-level-security policies, AES-256-GCM encryption for patient data, and full audit logging.
- Shipped two companion apps end-to-end: Retina Academy, a staff-training platform with video modules and quizzes, and a locked-down Android (Kotlin) clinic kiosk, and safeguarded releases with a Vitest, Playwright, and CI pipeline.

Software Engineer

May 2025 - Present

IllumibotAI

- Shipped the IllumibotAI consumer app in Flutter across iOS, Android, and web, including a self-built multi-timeline video editor with FFmpeg export and hardware-accelerated encoding that drove thousands in revenue and anchored a key partnership.
- Developed 40+ Node.js/TypeScript serverless functions orchestrating 10+ AI providers through event-triggered pipelines with atomic token transactions.
- Built native iOS and Android plugins with thermal monitoring and adaptive performance to prevent throttling during long video exports.

Founding Engineer

Feb 2024 - Dec 2024

OmniSynkAI

- Led a team of 5 to build a B2B e-commerce marketplace from 0 to 1, owning the backend and data layer end-to-end.
- Built a Node.js (Fastify), PostgreSQL, and Prisma backend that automates product listings, order fulfillment, and inventory sync across multiple marketplaces, deployed on GCP with GitHub Actions CI/CD.
- Cut backend response time 30% with Redis caching and fine-tuned a BERT model on 100K+ product entries to 95% accuracy for listing auto-categorization.

PROJECTS

Autonomous Job-Application Pipeline

- Built an async Python pipeline that discovers roles across LinkedIn, Greenhouse, and Lever via httpx, then filters, scores, and queues them in a WAL-mode SQLite database.
- Automated submissions with a Playwright and Gemini agent loop that extracts form fields, returns structured JSON actions, and executes them, orchestrated by APScheduler with per-platform rate limits and Discord alerts.

FreeFEM-CodeGen

- Built a two-stage QLoRA fine-tune (continued pretraining plus supervised fine-tuning) for generating FreeFEM++ finite-element code, lifting execution pass rate from 0% to 79.7% on held-out prompts, matching Gemini 3 Pro and beating 123B Mistral Large 2.1, trained on a single A100 in 67 minutes.

Sprig (Rust)

- Built a sparse n-gram indexed code-search engine in Rust that delivers sub-millisecond regex search across large codebases.

OPEN-SOURCE CONTRIBUTIONS

- [ByteDance trae-agent](#) (11k+ stars) — Merged 12 PRs adding Gemini and OpenAI (GPT-4o/o3) provider integrations and a JSONPath-based edit tool with 17 unit tests.
- [Unsloth](#) (28k+ stars) — Patched a mixed-precision training crash (FP16-to-FP32 embedding upcast) and a Gemma 3 vision processor bug.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Healthcare & Data: EHR Integration (Athena, Nextech), OAuth 2.0, HIPAA Compliance, PostgreSQL, Supabase, Drizzle ORM, Prisma

Backend: Node.js, Fastify, Express, REST APIs, Redis, FastAPI

Frontend & Tooling: React, Next.js, Flutter, Docker, GitHub Actions, Vitest, Playwright